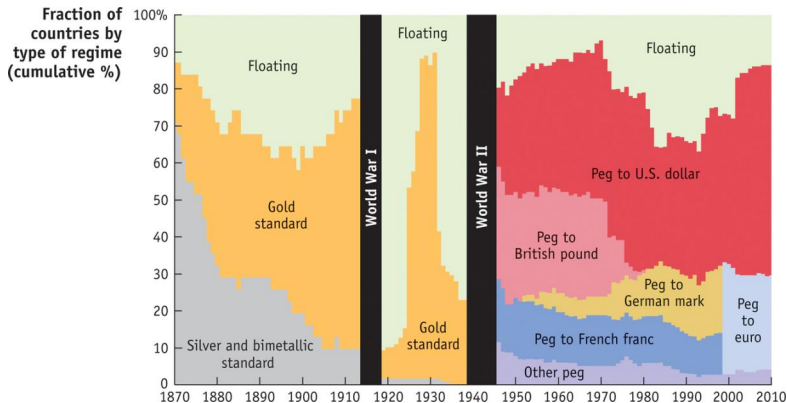


Macroeconomics

Exchange Rate Regimes

(Blanchard, Macroeconomics 8th edition Ch. 20)

Exchange Rate Regimes of the World



Gold Standard 1870–1914

- ▶ Each central bank fixed value of its currency relative to quantity of gold by trading domestic assets in exchange for gold
- ▶ No center country (i.e., symmetric)
- ▶ Countries did not peg x-rate at $\bar{E}$ (local currency price of some base currency), instead pegged at $\bar{P}_g$ (local currency price of gold)
- ▶ e.g., price of gold fixed at \$35 per ounce by Federal Reserve, 17.5 pounds per ounce by Bank of England $\Rightarrow$ dollar-pound x-rate must be fixed at \$2 per pound
- ▶ Gold standard rested on principle of free convertibility: central banks stood ready to buy and sell gold in exchange for paper money at these mint prices

Gold Standard 1870–1914 (con.)

- ▶ Gold standard also acted as an automatic restraint on increasing money supplies too quickly
 - ▶ Preventing inflationary monetary policies
- ▶ But restrained central banks from increasing money supply (i.e., monetary policy) to increase aggregate demand, output, and employment
- ▶ And price of gold relative to other goods and services varied, depending on the supply and demand of gold
 - ▶ New supply of gold made gold abundant (cheap), and prices of other goods rose because currency price of gold was fixed
 - ▶ Strong demand for gold jewelry made gold scarce (expensive), and prices of other goods fell because currency price of gold was fixed

Gold Exchange Standard and Bretton Woods

- ▶ Architects of post WWII order (e.g., Harry Dexter White, John Maynard Keynes) constructed a system keeping fixed x-rates but now with capital controls: encourage rebuilding of trade
 - ▶ Countries would peg to USD (center currency), which in turn, pegged to gold at fixed price
- ▶ Allows more flexibility in growth of international reserves, depending on macroeconomic conditions, because amount of currencies held as reserves could change
- ▶ Less constrained on supply and demand of gold

Bretton Woods to the Present

- ▶ Capital mobility grew, controls failed to hold: trilemma says that countries pegged to dollar stood to lose monetary policy autonomy
- ▶ Devaluation option came to be seen as most important way of achieving policy compromise in a “fixed but adjustable” system
- ▶ But increasingly frequent devaluations (and some revaluations) undermined notion of truly fixed rate system → more unstable
- ▶ US would eventually abandon its commitment to convert dollars into gold (August 1971)
- ▶ <https://www.imf.org/external/about/histend.htm>

After Bretton Woods

- ▶ Most advanced countries have opted to float and preserve monetary policy autonomy
- ▶ A group of European countries instead decided to try to preserve fixed exchange rate system among themselves (Euro)
- ▶ Some developing countries have maintained capital controls, but many (especially emerging markets) have opened capital markets
- ▶ Some countries, both developed and developing, have camped in the middle ground
 - ▶ Attempted to maintain intermediate regimes, such as “dirty floats” or pegs with “limited flexibility”

Exchange Rate Regimes

- ▶ As we have seen, during the Bretton Woods period between 1944 and 1972, a number of countries adopted a system of fixed x-rates.
- ▶ All member countries other than the US fixed the price of their currency in terms of dollar.
- ▶ Since then, the world has been characterized by many exchange rate arrangements.
- ▶ Which exchange rate regime should countries choose?
- ▶ There is no perfect exchange rate regime.
- ▶ Both fixed and flexible exchange rates have advantages and shortcomings.

The Medium Run

- ▶ Is a flexible exchange rate regime more attractive?
- ▶ With a fixed exchange rate regime, a country gives up two macroeconomic instruments: exchange rate and interest rate.
- ▶ In the medium run, the difference between the two regimes fades away.
- ▶ Economy reaches the same real exchange rate and the same level of output whether it operates under a fixed or flexible exchange rate.

The Medium Run (con.)

- ▶ The real exchange rate is:

$$\varepsilon = \frac{EP}{P^*}$$

- ▶ which can adjust either:
 - ▶ through a change in the nominal exchange rate, E :
 - ▶ If the domestic price level and foreign price level do not change in the short run, this is the only way to adjust the real exchange rate in the short run.
 - ▶ through a change in the domestic price level relative to the foreign price level:
 - ▶ In the medium run, as prices adjust, this option is open even to a country with a fixed nominal exchange rate.

Example

- ▶ There are three ways in which a US car can become cheaper relative to a Japanese car:
 1. Through a decrease in the dollar price of the US car
 2. Through an increase in the yen price of the Japanese car
 3. Through a decrease in the nominal exchange rate, i.e., a decrease in the value of the dollar in terms of the yen.

IS Relation under Fixed X-rates

- Recall the equation for goods-market equilibrium:

$$Y = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, \varepsilon)$$

- Also:

$$r = i - \pi^e$$

$$\varepsilon = \frac{EP}{P^*}$$

- Under fixed exchange rates, we must have:

$$E = \bar{E}$$

$$i = i^*$$

IS Relation under Fixed X-rates (con.)

- ▶ The IS relation in an open economy with fixed exchange rates:

$$Y = Y \left(\overset{\ominus}{\frac{\bar{E}P}{P^*}}, \overset{\oplus}{G}, \overset{\ominus}{T}, \overset{\oplus}{i^* - \pi^e}, Y^* \right)$$

- ▶ Demand and output depend:
 - ▶ negatively on the real exchange rate
 - ▶ positively on government spending and negatively on taxes
 - ▶ negatively on the domestic real interest rate
 - ▶ positively on foreign output through the effect on exports

Implications

- ▶ Recall the Phillips curve relation in the medium run:

$$\pi - \pi^e = (\alpha/L)(Y - Y_n)$$

- ▶ Assume constant expected inflation so that :

$$\pi - \bar{\pi} = (\alpha/L)(Y - Y_n)$$

- ▶ which implies that, if two economies are operating at potential, inflation rates would be the same, relative prices would remain constant, and so would the real exchange rate.

Implications (con.)

- ▶ In the medium run, the real exchange rate can adjust even if the nominal exchange rate is fixed.
- ▶ Consider a situation where output is below potential, then domestic inflation is lower than it would be if output were at potential, and thus lower than foreign inflation.
- ▶ This adjustment is achieved through movements in the relative price levels over time.
- ▶ For a given fixed nominal exchange rate, the real exchange rate decreases. Net exports increase over time, and so does output.
- ▶ In the medium run, output is back at potential, and domestic inflation and foreign inflation are equal. With domestic and foreign inflation being equal, the real exchange rate is constant.

Keynes versus Churchill

- ▶ The gold standard was a system in which each country fixed the price of its currency in terms of gold and stood ready to exchange gold for currency at the stated parity.
- ▶ In 1925, Britain decided to return to the gold standard, which was in place from 1870 until World War 1.
- ▶ As Keynes predicted, because the prices of British goods had increased more than the foreign price level, Britain faced a real appreciation at a given nominal exchange rate.

$$\varepsilon = \frac{EP \uparrow\uparrow}{P^* \uparrow}$$

Keynes versus Churchill (con.)

- ▶ Keynes severely criticized the decision to return to the pre-war parity.
- ▶ In *The Economic Consequences of Mr. Churchill*, a book he published in 1925, Keynes argued as follows:
- ▶ If Britain were going to return to the gold standard, it should have done so at a lower price of currency in terms of gold; that is, at a lower nominal exchange rate than the pre-war nominal exchange rate.
- ▶ Keynes's prediction turned out to be right. While other countries grew, Britain remained in a recession the rest of the decade.

Exchange Rate Crises under Fixed Exchange Rates

- ▶ Suppose a country has chosen to operate under a fixed exchange rate.
- ▶ Suppose also that financial investors start believing there may soon be an exchange rate adjustment— either a devaluation or a shift to a flexible exchange rate regime accompanied by a depreciation.
- ▶ As soon as financial markets believe a devaluation may be coming, then maintaining the exchange rate requires an increase—often a large one—in the domestic interest rate.
- ▶ To let a currency **float** is to allow a move from a fixed to a flexible exchange rate regime.

Moving from Fixed to Floating

- ▶ Given the interest rate condition:

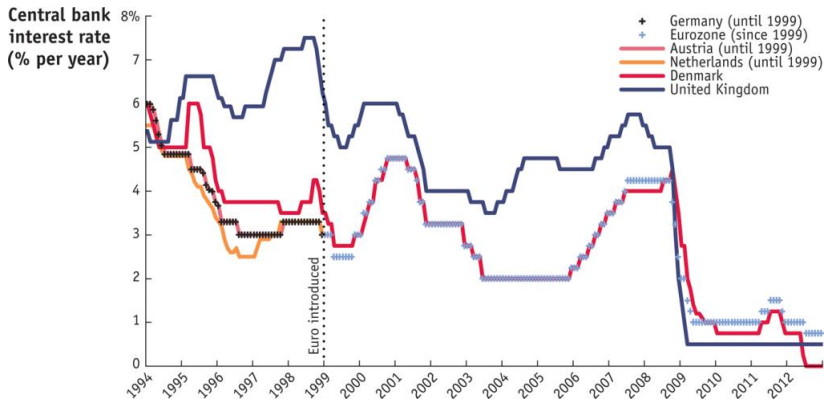
$$i = i_t^* - \frac{E_{t+1}^e - E_t}{E_t}$$

- ▶ which implies that as soon as financial markets believe a devaluation may be coming, maintaining the exchange rate requires an increase in the domestic interest rate.
- ▶ Expectations that a devaluation may be coming can trigger an exchange rate crisis.
- ▶ Faced with expectations of devaluation, the govt has 2 options:
 - ▶ Give in and devalue, or
 - ▶ Fight and maintain the parity, at the cost of high interest rates and a potential recession.

European Monetary System

- ▶ The European Monetary System (EMS) determined the movements of exchange rates within the European Union from 1978 and 1998.
- ▶ In the EMS, member countries agreed to maintain their exchange rate relative to the other currencies in the system within narrow limits or bands around a central parity—a given value for the exchange rate.
- ▶ Beginning on January 1, 1999, a number of those European countries adopted a common currency, the euro.

Trilemma in Europe



German Reunification, Interest Rates, and the EMS

- ▶ German reunification in 1990 led to large increases in demand and interest rates in Germany.
- ▶ To stay in the fixed exchange regime of the EMS, other European members had to match German interest rates, leading to decreases in their output.
- ▶ Under a fixed exchange rate regime such as the European Monetary System (EMS)—the system that prevailed before the introduction of the euro—no individual country can change its interest rate if the other countries do not change theirs as well.
- ▶ Germany changed its interest rate and other EMS countries were forced to match the interest rate.

German Reunification, Interest Rates, and the EMS

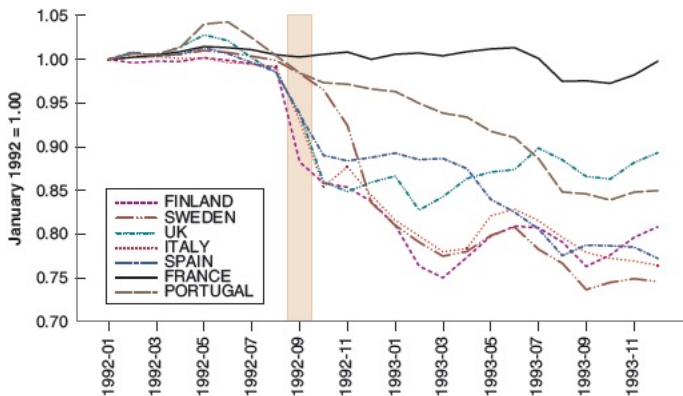
	<i>Nominal Interest Rates (%)</i>			<i>Inflation (%)</i>		
	1990	1991	1992	1990	1991	1992
Germany	8.5	9.2	9.5	2.7	3.7	4.7
France	10.3	9.6	10.3	2.9	3.0	2.4
Belgium	9.6	9.4	9.4	2.9	2.7	2.4
	<i>Real Interest Rates (%)</i>			<i>GDP Growth (%)</i>		
	1990	1991	1992	1990	1991	1992
Germany	5.8	5.5	4.8	5.7	4.5	2.1
France	7.4	6.6	7.9	2.5	0.7	1.4
Belgium	6.7	6.7	7.0	3.3	2.1	0.8

The 1992 EMS Crisis

- ▶ An example of the such a problem is the exchange rate crisis that shook the European Monetary System (EMS) in the early 1990s.
- ▶ In September 1992, financial markets became convinced realignments of the fixed exchange parities with bands were coming soon.
- ▶ Speculative attacks led to ensuring depreciations/devaluations of different countries.

The 1992 EMS Crisis

- Exchange Rates of Selected European Countries Relative to the Deutsche Mark, January 1992 to December 1993



Source: IMF database.

Black Wednesday and George Soros

- ▶ Changes in market sentiment, often called “animal spirits,” emerge from a model based on rational actors.
- ▶ In some circumstances, the instability problem is much worse than anticipated.
- ▶ One reason is that as the time frame shrinks, the currency premium (i.e., E^e/E or $\Delta E^e/E$) explodes. Another is how beliefs are formed in the first place.
- ▶ Investor George Soros likes to use the term “reflexivity” to describe how markets shape events, as well as vice versa:
- ▶ “Reflexivity is, in effect, a two-way feedback mechanism in which reality helps shape the participants’ thinking and the participants’ thinking helps shape reality.”

Black Wednesday and George Soros

- ▶ Shorting: Soros's firm borrowed billions of pounds and parked all the money in German mark deposits. "It was an obvious bet, a one-way bet," he later recalled.
- ▶ The Bank of England, under orders from the U.K. Treasury, was intervening furiously, selling marks to prop up the pound at the limits of the ERM band.
- ▶ The sudden increase in the currency premium was inducing a massive reserve drain. On the morning of September 16, 1992, the pressure on the pound became intense.
- ▶ It was all over by lunchtime, with the bulk of the reserves lost, and an estimated 4 billion pounds spent in a futile defense.
- ▶ <https://www.youtube.com/watch?v=WBZnau8Px5E&t=445s>
- ▶ <https://www.theguardian.com/business/2022/sep/13/black-wednesday-brexitester-sterling-crisis>

Flexible Exchange Rates

- ▶ Again, the interest parity condition:

$$1 + i_t = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right)$$

- ▶ Multiply both sides by E_{t+1}^e gives

$$E_t = \frac{1 + i_t}{1 + i_t^*} E_{t+1}^e$$

- ▶ Continue to solve forward in time for the expected exchange rate:

$$E_t = \frac{(1 + i_t)(1 + i_{t+1}^e) \dots (1 + i_{t+n}^e)}{(1 + i_t^*)(1 + i_{t+1}^{*e}) \dots (1 + i_{t+n}^{*e})} E_{t+n+1}^e$$

Flexible Exchange Rates (con.)

$$E_t = \frac{(1 + i_t)(1 + i_{t+1}^e) \dots (1 + i_{t+n}^e)}{(1 + i_t^*)(1 + i_{t+1}^{*e}) \dots (1 + i_{t+n}^{*e})} E_{t+n+1}^e$$

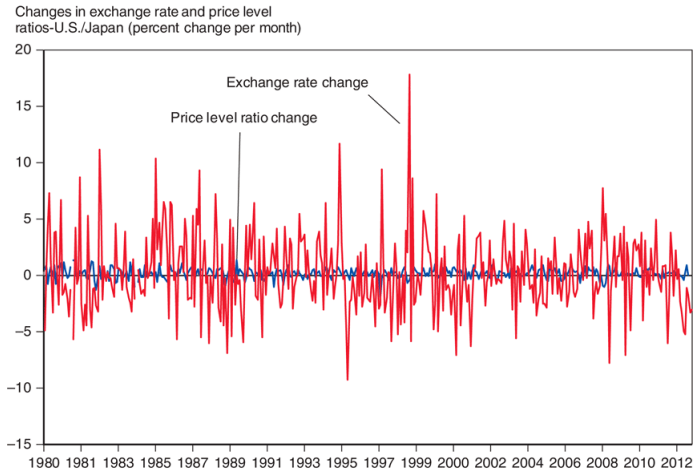
► Implications:

- The level of today's exchange rate will move one-for-one with the future expected exchange rate.
- Today's exchange rate will move when future expected interest rates move in either country.
- Because today's exchange rate moves with any change in expectations, the exchange rate will be volatile, i.e., move frequently and perhaps by large amounts.
- A country that decides to operate under flexible exchange rates must accept the fact that it will be exposed to substantial exchange rate fluctuations over time.

Exchange Rate Overshooting

- ▶ Exchange rate is said to *overshoot* when its immediate response to a change is greater than its long-run response
- ▶ Overshooting is predicted to occur when monetary policy has an immediate effect on interest rates, but not on prices and (expected) inflation
- ▶ Overshooting helps explain why exchange rates are so volatile

Variability of X-rates vs Prices



Source: Price levels from International Monetary Fund, *International Financial Statistics*. Exchange rate from Global Financial Data.

Lessons Learned

- ▶ At the end of the Bretton Woods period, countries moved from fixed exchange rates to flexible exchange rates.
- ▶ Most economists expected that exchange rates, while flexible, would be stable.
- ▶ The large fluctuations in exchange rates that followed—and have continued to this day—came as a surprise.
- ▶ For some time, these fluctuations were thought to be the result of irrational speculation in foreign exchange markets.
- ▶ It was not until the mid-1970s that economists realized that these large movements could be explained, by the rational reaction of financial markets to news about future interest rates and the future exchange rate.

Regime Choice: Integration and Similarity

- ▶ Why do some countries choose to fix and others to float?
- ▶ Why do they change their minds at different times?
- ▶ Fundamental source of divergence between what Britain and Germany wanted: each country faced different shocks
- ▶ Fiscal shock that Germany experienced after reunification was not felt in Britain or any other ERM country
- ▶ Issues that are at the heart of this decision are:
 1. *Economic integration*: as measured by trade and other transactions, and
 2. *Economic similarity*: as measured by the similarity of shocks

Economic Integration and Gains in Efficiency

- ▶ Economic integration refers to growth of market linkages in goods, capital, and labor markets among regions and countries
- ▶ By lowering transaction costs, a fixed exchange rate might promote integration and hence increase economic efficiency
- ▶ Why?
- ▶ The greater the degree of economic integration between markets in two countries, the greater will be the volume of transactions between the two
- ▶ Therefore, the greater will be the benefits the home country gains from fixing its exchange rate with the base country
- ▶ *As integration rises, efficiency benefits of a common currency increase*

Economic Similarity and costs of Asymmetric Shocks

- ▶ Fixed x-rate can be costly when a country-specific shock is not shared by the other country, i.e., shocks were dissimilar
- ▶ e.g., German policymakers wanted to tighten monetary policy to offset a boom, while
- ▶ British policymakers did not want to implement same policy (no such shock)
- ▶ For a home country that unilaterally pegs to a foreign country, asymmetric shocks impose costs in terms of lost output
- ▶ The greater the degree of economic similarity between home and base country (i.e., face more symmetric shocks and fewer asymmetric shocks), the smaller the economic stabilization costs to home of fixing its exchange rate to base
- ▶ *As economic similarity rises, stability costs of a common currency decrease*

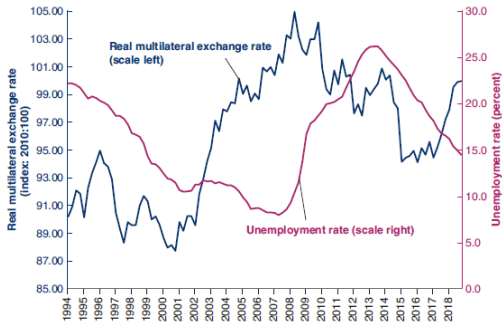
Symmetry-Integration Diagram

Choosing between Exchange Rates Regimes

- ▶ According to Robert Mundell, an optimal currency area needs to satisfy one of the three conditions:
 - ▶ The countries have to experience similar shocks.
 - ▶ Or, if the countries experience different shocks, prices and wages must be very flexible.
 - ▶ Or, if the countries experience different shocks, they must have high factor mobility.
- ▶ The common currency area composed of the 50 states of the US is close to an optimal currency area, which has high factor mobility, although different states are affected by different shocks.
- ▶ What about the eurozone?

Unemployment Rates in Spain

- ▶ A boom from 1994 to 2007 led to a steady decrease in unemployment, but also a steady real appreciation.
- ▶ In 2008, the boom turned into a bust, with a sharp increase in unemployment until 2013, and a real depreciation.



The Euro: A Short History

- ▶ As the European Union (EU) celebrated its 30th anniversary in 1988, a number of governments decided the time had come to plan a move to a common currency.
- ▶ They asked Jacques Delors, president of the EU, to prepare a report, which he presented in June 1989.
- ▶ The Delors report suggested moving to a European Monetary Union (EMU) in three stages.

The Euro: A Short History (con.)

- ▶ Stages moving to a European Monetary Union (EMU):
 - ▶ Stage 1: Beginning in 1990, capital controls were abolished.
 - ▶ Stage 2: Beginning in 1994, fixed parities of the exchange rate were maintained.
 - ▶ Stage 3: Beginning in 1999, parities between the currencies and a common currency, the euro, were “irrevocably” fixed; the new European Central Bank (ECB) became responsible for monetary policy for the Euro area.
- ▶ <https://www.youtube.com/watch?v=hJUWuYUvozC>

Hard Pegs

- ▶ To the extent that financial markets expect the parity to be maintained, they will stop worrying about money growth being used to finance the budget deficit.
- ▶ Fixing the exchange rate is not a magic solution.
- ▶ The country also needs to convince financial investors that not only is the exchange rate fixed today, it will remain fixed in the future.
- ▶ Ways to fix the exchange rate and convince financial investors that the exchange rate is fixed today and also in the future:
 - ▶ Make the fixed exchange rate be part of a more general macroeconomic package
 - ▶ Make a hard peg so that it is symbolically or technically harder to change the parity

Hard Pegs (con.)

- ▶ Examples of a hard peg:
 - ▶ Dollarization: Replace the domestic currency with the dollar
 - ▶ Currency board: A central bank stands ready to exchange foreign currency for domestic currency at the official exchange rate set by the government
- ▶ `https://www.imf.org/external/np/mfd/er/2004/eng/0604.htm`

Lessons from Argentina's Currency Board

- ▶ When Carlos Menem became President of Argentina in 1989, he inherited an economic mess.
- ▶ Inflation was running at more than 30% per month. Output growth was negative.
- ▶ Menem and his economy minister, Domingo Cavallo, quickly concluded that under these circumstances, the only way to bring money growth—and by implication, inflation—under control was to peg the peso (Argentina's currency) to the U S dollar and to do this through a hard peg (the currency board).
- ▶ In 1991, the Argentina's president of announced the adoption of a currency board that would stand ready to exchange its pesos for dollars on demand.

Lessons from Argentina's Currency Board (con.)

- ▶ The currency board appeared to work well for a while as inflation fell from 2,300% in 1990 to 4% in 1994.
- ▶ However, as appreciation of the dollar in the second half of the 1990s led to a decrease in demand for Argentina's goods, leading to declines in output.
- ▶ In early 2002, Argentina gave up the currency board and let the peso float.